

Phase-Field Crystal Framework for Single-Crystal Copper: Summary

Project Memo

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1 Background and Objectives

The project replicates the ductile-fracture framework of Xu *et al.*[1] to predict the tensile response of single-crystal copper (Cu) informed by EBSD observations. The primary objectives are: (i) build synthetic or EBSD-derived [111] Cu structures with defects, (ii) couple Phase-Field Crystal (PFC) density evolution with phase-field fracture and crystal plasticity, (iii) solve mechanical equilibrium for anisotropic FCC copper, evolve plasticity and cracks, and export results for Ovito/ParaView, and (iv) compare simulated crack paths, stress-strain curves, and fatigue metrics (Paris, Coffin-Manson) with experiments.

2 Energy Functional and Constraints

State variables include the PFC density ψ , crack field $\phi \in [0, 1]$, plastic strain ε^p , and grain/orientation markers η_g . The free energy reads

$$F = \int_{\Omega} [(1 - \phi)^2 f_{\text{elas}}^+ + f_{\text{elas}}^- + f_{\text{frac}} + f_{\text{grain}} + f_{\text{PFC}}] d\Omega. \quad (1)$$

Elastic energy employs the cubic stiffness tensor C_{ijkl} , rotated to the local orientation R via $C'_{ijkl} = R_{ip}R_{jq}R_{kr}R_{ls}C_{pqrs}$. Spectral decomposition of the strain tensor provides tensile/compressive splits: $\varepsilon = Q\Lambda Q^\top$, and Λ^+ keeps positive eigenvalues. Here ψ is the dimensionless PFC density deviation, ϕ is the damage/crack variable, ε^p is the plastic strain, η_g labels grain orientation, $f_{\text{elas}}^\pm$ are the positive/negative parts of the elastic energy density built from $\varepsilon - \varepsilon^p$, and f_{PFC} captures lattice-scale ordering.

Fracture energy

$$f_{\text{frac}} = G_c(\eta_g, \varepsilon_{\text{eq}}) \left[\frac{\phi^2}{2\ell_0} + \frac{\ell_0}{2} |\nabla \phi|^2 \right] \quad (2)$$

uses a toughness degradation law

$$G_c = G_{c0} \left(0.5 - g_{\text{res}} \tanh \left[2 \left(\frac{\varepsilon_{\text{eq}}}{\varepsilon_{1/2}} - 1 \right) \right] + 0.5 + g_{\text{res}} \right), \quad (3)$$

which lowers G_c with accumulated plastic strain. Parameters: G_{c0} initial toughness, ℓ_0 regularization length, g_{res} residual toughness ratio, and $\varepsilon_{1/2}$ the equivalent plastic strain at half-toughness. The equivalent plastic strain ε_{eq} is derived from the PFC density gradients or an explicit plastic field, matching the code path in **PFCCoupling**. Crack driving compares tensile elastic energy density with the critical fracture energy density G_c/ℓ_0 , so the numerical driving term uses $H\ell_0/(G_c + \epsilon)$; this reintroduces the characteristic length to maintain a dimensionless ratio and avoid blow-up.

Plastic surrogates and anisotropic driving Mechanical and PFC contributions are blended:

$$\hat{\varepsilon}_{\text{vm}} = \frac{\sqrt{1.5 \text{dev}(\varepsilon) : \text{dev}(\varepsilon)}}{\max |\cdot|}, \quad \hat{g} = \frac{\sqrt{\sum_i (\partial_i \psi)^2}}{\max |\cdot|}, \quad (4)$$

$$\varepsilon_{\text{eq}} = w_{\text{mech}} \hat{\varepsilon}_{\text{vm}} + (1 - w_{\text{mech}}) \hat{g}, \quad (5)$$

$$\varepsilon_{\text{dir},i} = w_{\text{mech}} [\varepsilon_{ii}]_+ + (1 - w_{\text{mech}}) \frac{|\partial_i \psi|}{\max_j |\partial_j \psi|}, \quad (6)$$

where $w_{\text{mech}} \in [0, 1]$ (default 0.9) and $[\cdot]_+$ clamps to tension. The loading-axis component $\varepsilon_{\text{dir},L}$ boosts both the history and crack driving:

$$H \leftarrow \max(H, f_{\text{elas}}^+ (1 + k_{\text{dir}} \varepsilon_{\text{dir},L})), \quad \mathcal{D} = \frac{H \ell_0}{G_c + \epsilon} (1 - \phi) (1 + k_{\text{dir}} \varepsilon_{\text{dir},L}), \quad (7)$$

with k_{dir} (default 0.8) tuning anisotropy. When stress coupling is enabled, the chemical potential gets an additive term

$$\mu_{\text{extra}} = \alpha \frac{\sigma_{\text{vm}}}{\max |\sigma_{\text{vm}}|}, \quad (8)$$

so high von Mises regions accelerate PFC evolution.

The PFC energy follows

$$f_{\text{PFC}} = \frac{\psi}{2} [r + (q_0^2 + \nabla^2)^2] \psi + \frac{u}{4} \psi^4, \quad (9)$$

with constraints (density conservation, grain fractions, plastic incompressibility) enforced through Lagrange multipliers in the modified TDGL formulation. Here r is the undercooling parameter that sets the ordering tendency, q_0 is the principal reciprocal lattice magnitude, and u penalizes large density excursions; the noise amplitude only seeds initial defects and does not enter the energy.

3 Mechanical Equilibrium

Small-strain equilibrium is imposed:

$$\nabla \cdot \boldsymbol{\sigma} = \mathbf{0}, \quad \boldsymbol{\sigma} = C'(\mathbf{x}) : (\varepsilon - \varepsilon^p), \quad (10)$$

with periodic in-plane boundaries and prescribed macroscopic strain $\varepsilon^{\text{macro}}$ in the loading direction. A conjugate-gradient linear operator solves for displacement $\mathbf{u}$, yielding total strain $\varepsilon = \nabla_s \mathbf{u} + \varepsilon^{\text{macro}}$. The rotated tensor $C'(\mathbf{x})$ comes from local orientation $R(\mathbf{x})$; ε^p is either driven by PFC gradients or provided explicitly.

4 Phase-Field Crystal Dynamics

PFC density obeys conserved dynamics:

$$\frac{\partial \psi}{\partial t} = -\nabla^2 \mu, \quad \mu = r\psi + (q_0^2 + \nabla^2)^2 \psi + u\psi^3, \quad (11)$$

implemented spectrally with FFTs to evaluate $(q_0^2 - k^2)^2$ and advance ψ semi-implicitly.

5 Crack Evolution

A history variable $H(\mathbf{x}) = \max_{t' \leq t} f_{\text{elas}}^+(\mathbf{x}, t')$ accumulates tensile energy. Crack driving force combines energy and toughness:

$$\mathcal{D} = \frac{H \ell_0}{G_c + \epsilon} (1 - \phi), \quad (12)$$

which compares the current tensile energy density to the critical fracture energy density G_c/ℓ_0 (here $\ell_0 \approx 3 \times 10^{-6}$ m). The evolution law $\dot{\phi} = L_\phi \mathcal{D}$ is clamped to $[0, 1]$, so cracks grow faster where energy is high and toughness is low. Here H stores the maximum tensile elastic energy density seen so far, ϵ is a small numerical residual stiffness, and L_ϕ is the kinetic coefficient for the crack field.

6 Numerical Workflow

1. **Structure generation:** ‘Cu111StructureBuilder’ constructs grids ($32 \times 32 \times 16$, spacing 5×10^{-7} m) with $[111]$ orientation and random defects for testing; EBSD data can replace the synthetic mask in future runs.
2. **Alternating solver:** The cyclic load path is triangular $0 \rightarrow +\varepsilon_{\text{max}} \rightarrow 0 \rightarrow -\varepsilon_{\text{max}} \rightarrow 0$, each segment subdivided into N_{seg} steps. At every increment: solve mechanical equilibrium (CG); blend mechanical/PFC plastic surrogates and relax plastic scalar + directional vector; update crack with anisotropic boost; advance PFC with optional stress-coupled μ_{extra} .
3. **Outputs:** Every few steps dump VTK frames; after each cycle dump LAMMPS and optional ‘data’ snapshots. Scalar/vector fields include crack, plastic (scalar + directional), ψ , stress (tensor and von Mises), and displacement. CSV logs store per-cycle energy, mean crack/plastic, crack increment, and plastic range; runs stop early if $\langle \phi \rangle$ exceeds a failure threshold.
4. **Fatigue metrics:** Fit crack increment per cycle vs cycle count (Paris-like slope) and plastic strain range vs $2N$ (Coffin–Manson slope) from the logged series.

7 Outlook

The framework replicates Fig. 1–4 behavior from Xu *et al.*[1] at the single-crystal scale. Next steps include (i) ingesting actual EBSD orientation/defect maps, (ii) swapping the CG equilibrium for FFT-based Lippmann–Schwinger or FEM for faster convergence and non-periodic BCs, and (iii) embedding full crystal-plasticity slip kinetics and validated fracture history laws.

参考文献

- [1] F. Xue *et al.*, “Phase-field framework with constraints and its applications to ductile fracture in polycrystals and fatigue,” *npj Computational Materials* 8, 18 (2022).

8 Recent Code Updates (Summary)

- 塑性与方向场：新增 `plastic_measures` 计算 ε_{eq} 与方向分量，机械 von Mises/轴向应变与 PFC 梯度按权重混合，默认机械占比 0.9。
- 方向性裂纹驱动：加载轴塑性正分量放大历史能量和驱动力 (`dir_coupling=0.8`)，抑制非加载方向裂纹。
- 应力耦合 PFC：von Mises 应力归一化后作为 μ_{extra} 加入化学势，提升高应力区的晶格演化响应。
- 求解与载荷：采用对称三角循环载荷 ($0 \rightarrow +\varepsilon_{max} \rightarrow 0 \rightarrow -\varepsilon_{max} \rightarrow 0$)，可设置失败阈值提前终止；跟踪每周期塑性范围 $\Delta\varepsilon_p$ 和裂纹增量。
- VTK/LAMMPS 输出扩展：输出塑性方向分量、应力张量/ von Mises，附带归一化场 (`crack_norm`, `plastic_vec_norm`, `stress_vm_norm`, `disp_total_norm` 等) 便于统一色标；VTK 现默认二进制 STRUCTURED_GRID (变形坐标)，避免 ASCII 读入报错，LAMMPS 去重了多余的 `plastic_vec` 列。
- 结构缺陷默认幅值调至 0.12，PFC 截断默认 $|\psi| \leq 1.2$ ，方便在高载下保持稳定。

9 Practical Simulation Recipe

1. **结构初始化**：用 `Cu111StructureBuilder` 构造网格（默认 [111] 单晶，或提供 grain labels/orientation map）。缺陷幅值默认 0.12，噪声注入 ψ ；可直接替换为 EBSD 导出的取向场。

2. **求解循环**: `run_virtual_cycles` 采用三角载荷 $0 \rightarrow +\varepsilon_{\max} \rightarrow 0 \rightarrow -\varepsilon_{\max} \rightarrow 0$, 默认每段 50 步, 可多周期。每步执行: 力学求解 (CG); 塑性标量 + 方向分量松弛; 裂纹更新 (方向增益); PFC 更新 (含应力耦合 μ_{extra})。
3. **参数提示**: `mech_plastic_weight=0.9`, `dir_coupling=0.8`, `plastic_relax=0.2`, `clip=1.2`, `failure_threshold=0.98`; 必要时调节应力耦合系数 (`mu_extra_from_stress`) 或加载轴 `load_axis`。
4. **输出与监控**: 每 5 步 VTK 帧 (`anim_frame_*.vtk`) 和每周期 LAMMPS dump (`virtual_cycle_*.lammprj`) CSV 日志记录能量、裂纹均值/增量、塑性均值/范围, 裂纹均值超阈值将提前停止。

10 Field/Output Guide

- **VTK frame files** (`sim/tests/virtual_cycle_vtk/anim_frame_*.vtk`): 二进制 STRUCTURED_GRID (若启用变形坐标), 含下列标量/向量; 加载时关闭 per-timestep rescale, 固定色标 (裂纹 0-0.3 或 0-1, 塑性/应力 0-1)。
- ψ : PFC 密度偏差, 红/蓝表示高/低密度或缺陷核; 受应力耦合影响会在高应力区演化。
- `crack`, `crack_norm`, `crack_clamp03`: 裂纹变量 (0-1), `clamp03` 便于观察萌生阶段, `norm` 便于固定色标。
- `plastic` (标量), `plastic_vec` (方向): 来自机械应变 + PFC 梯度; `plastic_vec_x_norm` 可直接用于加载方向色标, `plastic_vec_mag` 为模值。
- `stress_vm`, `stress_vm_norm`: von Mises 应力, 可用于判断高应力区; 亦用于 μ_{extra} 。
- `displacement`, `displacement_total`: 微观/总位移; `disp_total_norm` 提供归一化幅值, 避免色标跳变。
- **LAMMPS dump** (`sim/tests/virtual_cycle_0001.lammprj`): 列顺序为 `id`, `type`, `x`, `y`, `z`, `crack`, `plastic`, `psi`, `plastic_xx`, `plastic_yy`, `plastic_zz`, `stress_vm`, `stress_xx`, `stress_yy`, `stress_zz` (若场缺失则省略), 坐标包含宏 + 微观位移。
- **CSV 日志** (`sim/tests/virtual_cycle.csv`): `cycle` (循环编号)、`energy` (总能量)、`crack_mean` (裂纹均值)、`plastic_mean` (塑性均值)、`crack_delta` (该周期裂纹增量)、`plastic_range` (该周期塑性范围 $\Delta\varepsilon_p$)。
- 归一化场均在 VTK 中提供, 便于在 ParaView 固定色标 (range 0-1 或 0-0.3) 播放时间序列。

11 实验流程与数据对应

- **载荷与网格:** 默认单晶网格 $32 \times 32 \times 16$, 缺陷幅值 0.12; 三角循环载荷 ($0 \rightarrow +\varepsilon_{\max} \rightarrow 0 \rightarrow -\varepsilon_{\max} \rightarrow 0$), 每段 50 步, $\max_strain=0.08$, 可设置 $failure_threshold$ (0.98) 提前终止。
- **周期统计:** $energy$ 为周期末总能量; $crack_mean$ 代表损伤程度(接近 1 表示贯穿); $crack_delta$ 是相邻周期裂纹增量, 用于 Paris 拟合; $plastic_mean$ 为塑性均值; $plastic_range$ 为该周期塑性最大-最小 ($\Delta\varepsilon_p$ 替代量, 用于 Coffin-Manson 拟合)。
- **可视化要点:** 在 ParaView/Ovito 打开 VTK 或 LAMMPS dump 时, 优先查看 $crack_clamp03$ (萌生)、 $plastic_vec_x_norm$ (加载方向塑性)、 $stress_vm_norm$ (等效应力)、 $disp_total_norm$ (总位移)。固定色标 0-1 便于跨帧比较。
- **Coffin-Manson/Paris 指标:** 拟合对数斜率 $d \log(\Delta\varepsilon_p)/d \log(2N)$ 得到 Coffin-Manson; 拟合 $\log(\Delta\phi)/\log(N)$ 得到 Paris-like 系数, 均由 `sim/tests/virtual_cycle.py` 自动计算并打印。

12 当前结果快照

- **帧 1:** 裂纹 $\max \phi \approx 4.9 \times 10^{-7}$ (近零), 塑性 $\bar{\varepsilon}^p \approx 0.62$, $stress_vm$ 约 1.03×10^{-2} ; 宏观位移梯度 0-0.508 已写入 `displacement_total`。
- **帧 41 (循环中段):** 裂纹 $\phi \approx 0.014-0.018$, 塑性 0.84-0.94, 方向塑性模 0.24-0.36; 等效应力约 1.01×10^{-2} , 场较均匀。
- **帧 80 (循环末, 回到零应变):** 裂纹均值 $\approx 2.65 \times 10^{-2}$, 仍未贯穿; 塑性 0.72-0.82, 方向塑性接近 0; 应力/总位移回零, 符合卸载回零边界。
- **结论:** 当前载荷与韧性下仅出现轻微、均匀损伤和塑性累积, 无明显裂纹带或强应力集中; 若需更强局部化, 可提高载荷峰值/循环数、增大 `dir_coupling` 或降低韧性/提高缺陷幅值, 并保持固定色标判断真实局部化。